Hypothesis Testing with E-Values

Chapter 2: Validity – E-values under the Null

Based on Ramdas & Wang

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Chapter Overview

Outline

Notation You'll Need / Recap I

This chapter studies what happens to e-values *under the null hypothesis* $\mathcal{P}$ – the alternative $\mathbb{Q}$ never appears. Here is a compact, self-contained recap: a one-slide refresher on the Chapter 1 basics, plus the new ideas (calibrators, admissibility) that this chapter introduces.

Notation You'll Need / Recap II

Recap: e-variables, p-variables, and tests (from Chapter 1)

- **Null hypothesis** $\mathcal{P}$: a set of candidate probability distributions, one of which is assumed to be the "true" data-generating law if H_0 holds. Think of $\mathcal{P}$ as "all the ways the world could look if H_0 is true."
- **E-variable (e-value)** E **for** $\mathcal{P}$: a nonnegative random variable with $\mathbb{E}^{\mathbb{P}}[E] \leq 1$ for every $\mathbb{P} \in \mathcal{P}$. Intuition: a "bet" against H_0 that is fair or sub-fair on average whenever H_0 is true; a large realized value is evidence against H_0.
- **Exact e-variable**: one with $\mathbb{E}^{\mathbb{P}}[E] = 1$ for every $\mathbb{P} \in \mathcal{P}$ (no money left on the table, on average).
- **P-variable (p-value)** P **for** $\mathcal{P}$: a random variable with $\mathbb{P}(P \leq u) \leq u$ for all $u \in [0, 1]$ and all $\mathbb{P} \in \mathcal{P}$. Intuition: under H_0, P is "at least as spread out as uniform," so small values of P are surprising.
- **Exact p-variable**: P is exactly $\mathrm{Uniform}[0, 1]$ under every $\mathbb{P} \in \mathcal{P}$.
- **Level-α test**: a $[0, 1]$-valued random variable (a "rejection probability") whose $\mathbb{P}$-expectation is at most α for all $\mathbb{P} \in \mathcal{P}$; a *binary* test additionally only takes values in $\{0, 1\}$ (reject / do not reject).
- $\mathbb{P} \in \mathcal{M}_1$: shorthand for "$\mathbb{P}$ is a probability measure on the sample space," i.e., any legitimate candidate law.

Outline

2.1 Testing Using Markov's Inequality I

E-values can already be interpreted as evidence against H_0 in their own right. But it is natural to ask how an e-value can be converted – perhaps with some loss of efficiency – into the more classical machinery of level-α tests and p-values. The bridge is **Markov's inequality**, a two-line argument that turns *any* e-value into *any* desired test.

Proposition 2.1: Markov's Inequality for E-Values

Let E be an e-variable for $\mathcal{P}$. Then

$$\mathbb{P}(E \geq 1/\alpha) \leq \alpha \quad \text{for all } \mathbb{P} \in \mathcal{P},\ \alpha \in (0,1].$$

Hence: $1/E$ is a p-variable; $(\alpha E) \wedge 1$ is a level-α test; and $\mathbf{1}_{\{E \geq 1/\alpha\}}$ is a level-α binary test.

- E: the e-variable, a nonnegative random quantity with $\mathbb{E}^{\mathbb{P}}[E] \leq 1$ under H_0.
- α: the significance level you choose in advance (e.g. 0.05).
- $1/\alpha$: the e-value threshold – reject H_0 once E crosses it.
- $x \wedge 1 := \min(x, 1)$: clips a quantity at 1 so it is a valid "rejection probability."
- $\mathbf{1}_{\{A\}}$: the indicator, equal to 1 if event A happens and 0 otherwise.

2.1 Testing Using Markov's Inequality II

Proof of Proposition 2.1

Step 1. Start from the elementary inequality $\mathbf{1}_{\{x \geq 1\}} \leq x$ for all $x \geq 0$ (a jump function is bounded by the identity line once you check $x < 1$ and $x \geq 1$ separately).

Step 2. Substitute $x = \alpha E$: $\mathbf{1}_{\{\alpha E \geq 1\}} \leq \alpha E$, i.e., $\mathbf{1}_{\{E \geq 1/\alpha\}} \leq \alpha E$.

Step 3. Take $\mathbb{E}^{\mathbb{P}}[\cdot]$ of both sides (expectation preserves $\leq$):

$$\mathbb{P}(E \geq 1/\alpha) = \mathbb{E}^{\mathbb{P}}\left[\mathbf{1}_{\{E \geq 1/\alpha\}}\right] \leq \alpha\, \mathbb{E}^{\mathbb{P}}[E] \leq \alpha,$$

using $\mathbb{E}^{\mathbb{P}}[E] \leq 1$ in the last step (definition of e-variable). ∎

2.1 Testing Using Markov's Inequality III

Why This Is Conservative

Even though $1/E$ is technically a p-variable, it is usually a *bad* one: it cannot be uniformly distributed on $[0,1]$ (so it is never exact), and it is typically far from it. Equality in Markov's inequality only occurs when E is a two-point random variable supported on $\{0, 1/\alpha\}$ with mean 1 – a very special case. In general, the type-I error of $\mathbf{1}_{\{E \geq 1/\alpha\}}$ is strictly smaller than the nominal α.

2.1 Testing Using Markov's Inequality IV

Running Example: Applying Markov's Inequality to the Coin

We test $H_0 : \theta = 0.5$ using $n = 20$ flips and observe $x = 14$ heads. As our e-variable, take the likelihood ratio against the best-supported alternative $\theta_1 = x/n = 0.7$ (the maximum likelihood estimate):

$$E = \frac{\theta_1^x (1-\theta_1)^{n-x}}{\theta_0^x (1-\theta_0)^{n-x}} = \frac{0.7^{14} \times 0.3^6}{0.5^{20}} \approx 5.18.$$

(The binomial coefficient $\binom{n}{x}$ cancels between numerator and denominator.) This $E \approx 5.18$ is an exact e-variable for the simple null $\{\theta_0 = 0.5\}$ (Example 2.8 / Section 2.6.1 explain why).

Applying Proposition 2.1: solving $1/\alpha = E$ gives $\alpha = 1/E \approx 0.193$. So the test "reject if $E \geq 1/\alpha$" is only guaranteed level $\alpha \approx 0.193$ here – not very stringent, illustrating that Markov's inequality can be quite loose for a single, moderately-sized e-value.

2.1 Testing Using Markov's Inequality V

On the Book's Figure 2.2 (Redrawn Independently)

For the two-sided normal test of Chapter 1, comparing e-values (several δ) against the classical p-value on a $\log_{10}$ scale shows that it is *harder* for an e-value to reach 100 than for a p-value to reach 0.01 – direct visual evidence that the Markov bound is generally conservative. See Figure 1 below (own redraw).

2.1 Testing Using Markov's Inequality VI

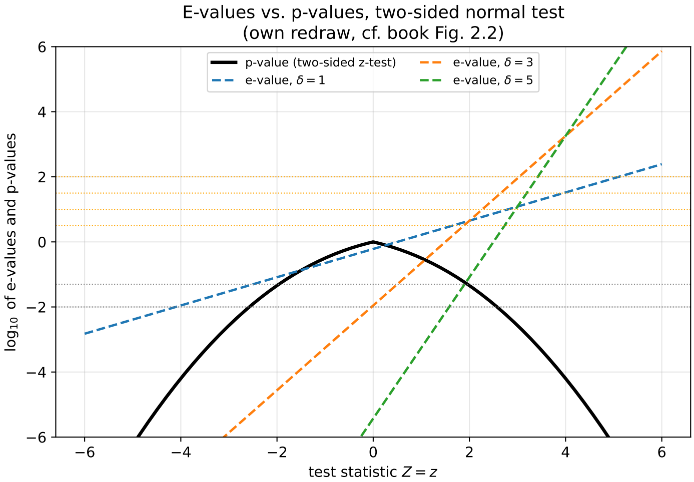
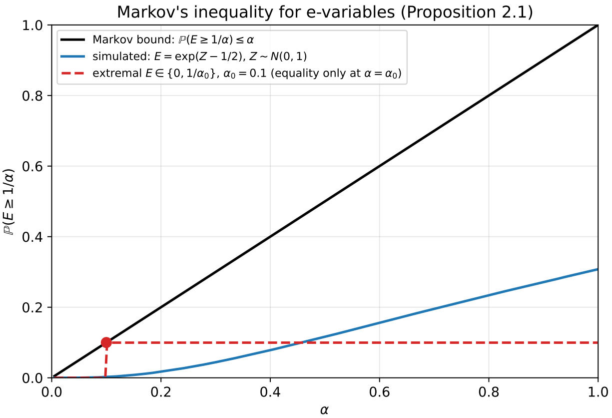


Figure: E-values (for several δ) vs. the p-value, two-sided normal test, as functions of the test statistic Z.

Figure: Markov's Inequality, Simulated



Reading the Plot

- Black line: the Markov bound $\alpha \mapsto \alpha$ itself (an upper bound on $\mathbb{P}(E \geq 1/\alpha)$).
- Blue curve: the *actual* tail probability for a simulated exact e-variable $E = \exp(Z - 1/2)$, $Z \sim N(0, 1)$ – always strictly below the bound (conservative).
- Red dashed: a two-point e-variable achieving *equality* exactly at $\alpha_0 = 0.1$, the only case where Markov's inequality is tight.

Python: Markov's Inequality on the Coin Example

```python
import numpy as np

n, x = 20, 14                       # n flips, x heads observed
theta0, theta1 = 0.5, x / n         # null vs. MLE alternative

# Likelihood-ratio e-value for H0: theta = theta0
loglik = lambda th: x*np.log(th) + (n-x)*np.log(1-th)
E = np.exp(loglik(theta1) - loglik(theta0))
print(f"e-value E = {E:.4f}")                          # 5.1844

# Markov's inequality: alpha = 1/E gives the guaranteed level
alpha_markov = 1 / E
print(f"Markov level alpha = 1/E = {alpha_markov:.4f}")    # 0.1929

# Monte Carlo check of Prop. 2.1 under the null: P(E>=1/alpha) <= alpha
rng = np.random.default_rng(0)
sims = rng.binomial(n, theta0, size=2_000_000)
def e_value(k):
    ll = lambda th: k*np.log(th) + (n-k)*np.log(1-th)
    return np.exp(ll(theta1) - ll(theta0))
E_all = e_value(sims)
alpha = 0.193
print("empirical P(E>=1/alpha):", np.mean(E_all >= 1/alpha), " <= alpha?", alpha)
```

Outline

2.2 Testing Using Distributional Knowledge I

Markov's inequality only uses the fact that $\mathbb{E}^{\mathbb{P}}[E] \leq 1$. If we actually know (or can compute) the full distribution of E under the null, we can find sharper, smaller thresholds – exactly as with any other classical test statistic.

Survival Function

For an e-variable E and $\mathbb{P} \in \mathcal{P}$, define the left-continuous survival function

$$S(x; \mathbb{P}) = \mathbb{P}(E \geq x), \qquad x \in \mathbb{R}.$$

If $S(\cdot\,; \mathbb{P})$ is the same for every $\mathbb{P} \in \mathcal{P}$, we call E **pivotal** (its null distribution does not depend on which $\mathbb{P}$ in $\mathcal{P}$ actually holds – a very convenient property, revisited in Section 14.3).

- $S(x; \mathbb{P})$: the chance, under a specific candidate law $\mathbb{P}$, that the e-value is at least x. This is exactly the p-value you would report if you used E itself (thresholded at x) as your test statistic.
- Pivotal: "the same regardless of nuisance parameters within H_0."

2.2 Testing Using Distributional Knowledge II

Sharper Thresholds

For a threshold $t > 0$, the test $\mathbf{1}_{\{E \geq t\}}$ has level α as soon as $\sup_{\mathbb{P} \in \mathcal{P}} S(t; \mathbb{P}) \leq \alpha$. Choosing $t \geq 1/\alpha$ always works (Markov), but we can typically do better. If $\mathbb{P} \mapsto S(t; \mathbb{P})$ has a continuous supremum, the *smallest* valid threshold is

$$t^* = \inf \left\{ t > 0 \; : \; \sup_{\mathbb{P} \in \mathcal{P}} S(t; \mathbb{P}) \leq \alpha \right\}.$$

This t^* equals the largest left α-quantile of E over $\mathbb{P} \in \mathcal{P}$. When $\mathcal{P} = \{\mathbb{P}\}$ is simple, t^* reduces to the plain left α-quantile of E under $\mathbb{P}$.

2.2 Testing Using Distributional Knowledge III

Running Example: An Exact Threshold for the Coin

Reducing the threshold below $1/\alpha$ is just business as usual: treat E as an ordinary test statistic. For our coin e-value E (a function of $X =$ number of heads under $\theta_0 = 0.5$), we can compute the exact null distribution of E by evaluating $E(k)$ for every $k = 0, 1, \ldots, 20$, then read off the exact α-quantile t^* directly from the (fully known, discrete) Binomial(20, 0.5) law – typically much smaller than the crude Markov threshold $1/\alpha$. This is exactly what the Python cell in Section 2.1 already computes when it enumerates `e_value(k)` over all k.

Caveat

Even when $S(\cdot\,;\mathbb{P})$ is hard to pin down exactly, partial information about the distribution of E can still yield useful bounds on t^* that beat naive Markov – this is developed further in Chapter 15.

Outline

2.3 Calibrators: Definitions I

P-values and e-values can be converted into each other via fixed, universal "currency-exchange" functions called **calibrators**. The key requirement is that the conversion must work *for any* hypothesis – we are not allowed to peek at $\mathcal{P}$ when designing f.

Definition 2.3: Calibrators

(i) A **p-to-e calibrator** is a decreasing function $f : [0, \infty) \to [0, \infty]$ with $f = 0$ on $(1, \infty)$, such that for *any* hypothesis, $f(P)$ is an e-variable whenever P is a p-variable.

(ii) An **e-to-p calibrator** is a decreasing function $g : [0, \infty) \to [0, \infty)$ such that for any hypothesis, $g(E)$ is a p-variable whenever E is an e-variable.

(iii) Calibrator f **dominates** g if $f \geq g$ (p-to-e case) or $f \leq g$ (e-to-p case); domination is **strict** if further $f \neq g$. A calibrator is **admissible** if no other calibrator strictly dominates it.

- Decreasing: bigger p-value (weaker evidence) $\Rightarrow$ smaller e-value, and vice versa – calibrators must preserve the direction of evidence.
- $f = 0$ on $(1, \infty)$: p-values larger than 1 do not occur, so we just define f to vanish there for tidiness.

2.3 Calibrators: Definitions II

- Domination: "f never gives a worse exchange rate than g, and sometimes gives a strictly better one."
- Admissible: the best you can do – no other calibrator is uniformly at least as favorable.
- We often drop "p-to-e" and just say "calibrator," because (as we will see) e-to-p calibrators turn out to be essentially unique.
- **Atomless** $\mathbb{P}$: one under which some continuously distributed (e.g. uniform, normal) random variable exists. It suffices to verify calibrator properties on one atomless $\mathbb{P}$.

2.3 Calibrators: Definitions III

Proposition 2.4: The Unique E-to-P Calibrator

The function $f(t) = \min(1, 1/t)$ is an e-to-p calibrator. It dominates *every* other e-to-p calibrator. In particular, it is the *only* admissible e-to-p calibrator.

Proof Sketch

Step 1. $f(t) = \min(1, 1/t)$ is an e-to-p calibrator: this is exactly Markov's inequality (Proposition 2.1), giving $\mathbb{P}(1/E \leq u) \leq u$ when $u = 1/t$.

Step 2. Suppose g is another e-to-p calibrator with $g(t) < \min(1, 1/t)$ for some t. Fix an atomless $\mathbb{P}$. Two cases: (a) if $t > 1$ and $g(t) < 1/t$, build E equal to t with probability $1/t$ and 0 otherwise – then $g(E)$ fails to be a valid p-variable; (b) if $t \leq 1$ and $g(t) < 1$, the deterministic e-variable $E = t$ shows $g(E) = g(t) < 1$ is not uniform-dominated. Either way, g cannot be a calibrator, a contradiction. So no calibrator can beat $\min(1, 1/t)$ anywhere. ■

2.3 Calibrators: Definitions IV

Proposition 2.5: Characterizing P-to-E Calibrators

A decreasing $f : [0, \infty) \to [0, \infty]$ with $f = 0$ on $(1, \infty)$ is a p-to-e calibrator if and only if $\int_0^1 f(p)\,\mathrm{d}p \leq 1$. It is **admissible** if and only if f is upper semicontinuous (equivalently, left-continuous), $f(0) = \infty$, and $\int_0^1 f(p)\,\mathrm{d}p = 1$.

- $\int_0^1 f(p)\,\mathrm{d}p$: the average value of f applied to a Uniform$[0, 1]$ random variable – this must be at most 1 because $f(P)$ must itself be a valid e-variable (mean ≤ 1).
- $f(0) = \infty$: an admissible calibrator must reward a p-value of exactly 0 with infinite evidence – otherwise it can be improved.
- Unlike the e-to-p direction, the class of admissible p-to-e calibrators is *much richer* – there is no single best choice.

Proof Idea ("if" direction)

Step 1. Since f is decreasing and $P' \sim \mathrm{Uniform}[0,1]$ stochastically dominates any p-variable P (i.e. $\mathbb{P}(P \le x) \le \mathbb{P}(P' \le x)$), we get $\mathbb{P}(f(P) > x) \le \mathbb{P}(f(P') > x)$ for all x.

Step 2. Integrating tail probabilities gives $\mathbb{E}^{\mathbb{P}}[f(P)] \le \mathbb{E}^{\mathbb{P}}[f(P')] = \int_0^1 f(p)\,\mathrm{d}p \le 1$, so $f(P)$ is indeed an e-variable. ■

2.3 Calibrators: Definitions VI

A Menu of Admissible P-to-E Calibrators

$$f(p) = \kappa\, p^{\kappa-1} \quad \text{for some } \kappa \in (0,1) \qquad \text{(power family, Eq. 2.1)}$$

$$f(p) = \int_0^1 \kappa p^{\kappa-1}\, \mathrm{d}\kappa = \frac{1-p+p\log p}{p(-\log p)^2} \qquad \text{(mixture, Eq. 2.2)}$$

$$f(p) = 2(1-p) \qquad \text{(Eq. 2.3)}$$

$$f(p) = p^{-1/2} - 1 \qquad \text{(Eq. 2.4)}$$

$$f(p) = -\log p \qquad \text{(Eq. 2.5)}$$

- κ (kappa): a free tuning parameter trading off how much evidence very small p gets versus moderate p.
- All five formulas above are admissible (Proposition 2.5): each integrates to exactly 1 over $p \in (0,1]$, is left-continuous, and blows up to ∞ as $p \to 0$.

2.3 Calibrators: Definitions VII

- Eqs. 2.3–2.5 penalize very small p-values *less* steeply than Eq. 2.2, so they are less likely to manufacture huge e-values, but can be more useful when multiplying many independent e-values together.

2.3 Calibrators: Definitions VIII

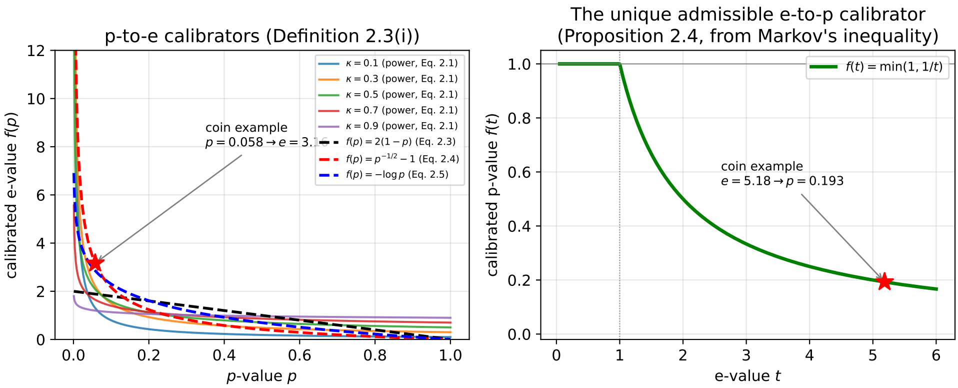


Figure: Left: several p-to-e calibrators, with the running coin example marked. Right: the unique admissible e-to-p calibrator $f(t) = \min(1, 1/t)$, same example marked.

2.3 Calibrators: Definitions IX

Running Example: Calibrating the Coin, Both Directions

Recall from Section 2.1: the coin data give a one-sided p-value $P \approx 0.0577$ (probability, under $\theta_0 = 0.5$, of 14 or more heads out of 20) and a likelihood-ratio e-value $E \approx 5.18$.

P-to-e, using Eq. 2.4: $f(p) = p^{-1/2} - 1$ gives

$$f(0.0577) = 0.0577^{-1/2} - 1 \approx 4.164 - 1 = 3.16.$$

This is a valid (if somewhat arbitrary) e-value manufactured purely from the p-value – notably *smaller* than the "natural" likelihood-ratio e-value 5.18 computed directly from the model.

E-to-p, using Proposition 2.4: $f(t) = \min(1, 1/t)$ applied to $E \approx 5.18$ gives

$$\min(1, 1/5.18) \approx 0.193,$$

a p-value much larger (more conservative) than the "natural" p-value 0.0577 – again showing the Markov conversion is lossy.

2.3 Calibrators: Definitions X

A Warning: Do Not Round-Trip

Converting p $\to$ e $\to$ p (or e $\to$ p $\to$ e) generally loses a lot of evidence. Book example: starting from $p = 0.01$, the calibrator $p \mapsto p^{-1/2} - 1$ gives $e = 9$; converting back with $e \mapsto \min(1/e, 1)$ gives $p' = 1/9 \approx 0.111$ – eleven times weaker than the original $p = 0.01$! **Moral**: calibrate once, in the direction you need, and stop.

Proposition 2.6 (Rescaling Calibrators)

If f is a (admissible) calibrator, so is $x \mapsto a\, f(ax)$ for any $a \geq 1$. *Proof idea*: a change of variables shows the integral constraint $\int_0^1 g(x)\,\mathrm{d}x = 1$ is preserved, and upper-semicontinuity / the value at 0 carry over directly from f.

2.3 Calibrators: Definitions XI

Proposition 2.7 (Every E-Variable Is a Calibrated P-Value)

Let E be an e-variable for an atomless $\mathbb{P}$. Then there exists a calibrator f and an exact p-variable P for $\mathbb{P}$ such that $E = f(P)$ $\mathbb{P}$-almost surely. If E is exact, f can be chosen admissible. *Proof idea*: take f to be (a version of) the right-quantile function of E; the standard quantile transform then produces an exact p-variable P with $E = f(P)$.

Example 2.8: The Two-Sided Normal Test

For $\mathbb{P} = N(0,1)$ vs. $\mathbb{Q} = N(\mu, 1)$ with n observations (Chapter 1's running example), the likelihood-ratio e-value is $E = \exp(\delta Z - \delta^2/2)$ and the Neyman–Pearson p-value is $P = \Phi(-Z)$, where $Z \sim N(0,1)$ under $\mathbb{P}$, Φ is the standard normal CDF, and $\delta = \mu\sqrt{n}$. Both are exact, and indeed $E = f(P)$ with the admissible calibrator

$$f(p) = \exp\big(-\delta\,\Phi^{-1}(p) - \delta^2/2\big).$$

Python: Calibrating the Coin Example

```python
import numpy as np
from scipy import stats

n, x = 20, 14
theta0 = 0.5
p_val = 1 - stats.binom.cdf(x - 1, n, theta0)      # one-sided p-value
print(f"p-value = {p_val:.4f}")                     # 0.0577

# p-to-e calibrator, Eq. (2.4): f(p) = p^{-1/2} - 1
e_from_p = p_val ** (-0.5) - 1
print(f"calibrated e-value f(p) = {e_from_p:.3f}")     # 3.16

# natural likelihood-ratio e-value (theta1 = MLE = 0.7)
theta1 = x / n
loglik = lambda th: x*np.log(th) + (n-x)*np.log(1-th)
E_natural = np.exp(loglik(theta1) - loglik(theta0))
print(f"natural e-value = {E_natural:.3f}")            # 5.18

# e-to-p calibrator (Proposition 2.4): f(t) = min(1, 1/t)
p_from_e = min(1, 1 / E_natural)
print(f"calibrated p-value min(1,1/E) = {p_from_e:.3f}")  # 0.193

# Warning: round-tripping loses evidence -- check with p=0.01
p0 = 0.01
e0 = p0 ** (-0.5) - 1                          # 9.0
p0_back = min(1, 1 / e0)                       # 0.111 (weaker than 0.01!)
print(f"round-trip: p={p0} -> e={e0:.2f} -> p'={p0_back:.3f}")
```

Outline

2.4 Randomized Tests and Calibrators I

A **randomized test** uses independent external randomness (e.g., an auxiliary coin flip you generate yourself) to sharpen a test. It turns out external randomization can strictly improve Markov's inequality, giving a strictly more powerful e-to-p calibrator.

Theorem 2.9: Randomized Markov Inequality

Fix $\mathbb{P} \in \mathcal{M}_1$. Let $Y \geq 0$ be a random variable and $U \sim \mathrm{Uniform}[0,1]$ independent of Y. Then for any $\alpha > 0$,

$$\mathbb{P}(Y \geq U/\alpha) = \mathbb{E}^{\mathbb{P}}[\min(\alpha Y, 1)] \leq \alpha\, \mathbb{E}^{\mathbb{P}}[Y].$$

In particular, if Y is an e-variable for $\mathcal{P}$, then $\mathbb{P}(Y \geq U/\alpha) \leq \alpha$ for all $\mathbb{P} \in \mathcal{P}$.

- U: an independent auxiliary uniform random draw – pure external randomness, generated by you, not by nature.
- Setting $U \equiv 1$ recovers ordinary Markov's inequality exactly, so Theorem 2.9 strictly generalizes Proposition 2.1.

2.4 Randomized Tests and Calibrators II

- If Y is bounded, $Y \in [0, C]$ a.s., the inequality becomes an *equality* for any $\alpha \leq 1/C$ – randomization can make the bound tight in cases where plain Markov cannot.

Proof

Step 1. Since U is independent of Y and uniform, conditioning on Y gives $\mathbb{P}(U \leq \alpha Y \mid Y) = \min(\alpha Y, 1)$.

Step 2. Taking total expectation: $\mathbb{P}(Y \geq U/\alpha) = \mathbb{E}^{\mathbb{P}}[\min(\alpha Y, 1)] \leq \alpha \mathbb{E}^{\mathbb{P}}[Y]$, using $\min(a, 1) \leq a$. ∎

2.4 Randomized Tests and Calibrators III

Corollary 2.10

For any hypothesis $\mathcal{P}$: if a p-variable P and an e-variable E are *independent*, then P/E is a p-variable. In particular, if E is an e-variable, then U/E is a p-variable, where $U \sim \mathrm{Uniform}[0,1]$ is independent of E.

Why Randomize?

Since $U < 1$ almost surely, U/E is a *strictly smaller* (hence more powerful) p-value than $1/E$ in all but degenerate cases. The map $e \mapsto (U/e) \wedge 1$ is itself an admissible *randomized* e-to-p calibrator (Section 11.3) that dominates the non-randomized $e \mapsto (1/e) \wedge 1$ – the latter being admissible only *among non-randomized* calibrators (Proposition 2.4).

2.4 Randomized Tests and Calibrators IV

Running Example: Randomizing the Coin Test

With $E \approx 5.18$, draw an independent $U \sim \mathrm{Uniform}[0, 1]$, say $u = 0.4$ (a value you generate yourself, unrelated to the coin data). Then the randomized p-value is

$$u/E \approx 0.4/5.18 \approx 0.077,$$

smaller (more powerful) than the non-randomized $1/E \approx 0.193$ from Section 2.3 – though of course a *different* draw of U gives a different randomized p-value, which is the price of randomization.

Outline

2.5 Admissible E-Variables I

Intuitively, an e-variable is **admissible** if it cannot be made “bigger” without breaking the e-variable property – no evidence is left on the table. This section characterizes admissibility precisely and connects it to likelihood ratios.

Definition 2.11: Admissibility

An e-variable E for $\mathcal{P}$ is **admissible** if for any other e-variable E' for $\mathcal{P}$ with $E' \geq E$, we have $\mathbb{P}(E' = E) = 1$ for every $\mathbb{P} \in \mathcal{P}$. Equivalently: E is admissible if it is not **dominated**, where E' dominates E means $E' \geq E$ always, and $\mathbb{P}(E' > E) > 0$ for some $\mathbb{P} \in \mathcal{P}$.

- Domination: E' is at least as large everywhere, and strictly larger with positive probability somewhere – so E' is a strict improvement on E.
- The constant 1 is a (trivial) admissible e-variable for any $\mathcal{P}$ – it cannot be improved because any $E' \geq 1$ dominating it would need $\mathbb{E}^{\mathbb{P}}[E'] > 1$ for some $\mathbb{P}$, violating the e-variable constraint.

2.5 Admissible E-Variables II

2.5.1 Admissible E-Variables for a Simple Null Are Likelihood Ratios

Proposition 2.12. For a simple null $\{\mathbb{P}\}$, an e-variable E is admissible if and only if it is exact ($\mathbb{E}^{\mathbb{P}}[E] = 1$). Further, every exact e-variable equals a likelihood ratio $\mathrm{d}\mathbb{S}/\mathrm{d}\mathbb{P}$ almost surely, for some distribution $\mathbb{S} \ll \mathbb{P}$ (absolutely continuous with respect to $\mathbb{P}$).

- $\mathrm{d}\mathbb{S}/\mathrm{d}\mathbb{P}$: the likelihood ratio (Radon–Nikodym derivative) of a candidate "evidence distribution" $\mathbb{S}$ against the null $\mathbb{P}$ – exactly the classical statistical likelihood ratio.
- $\mathbb{S} \ll \mathbb{P}$ ("$\mathbb{S}$ absolutely continuous with respect to $\mathbb{P}$"): $\mathbb{S}$ assigns zero probability to every event that $\mathbb{P}$ assigns zero probability to, so the ratio is well defined.

2.5 Admissible E-Variables III

Proof Idea

Step 1. If $\mathbb{E}^{\mathbb{P}}[E] < 1$, then $E + (1 - \mathbb{E}^{\mathbb{P}}[E])$ is an e-variable strictly dominating E – so non-exact e-variables are never admissible.

Step 2. Conversely, if E' dominates an exact E, then $\mathbb{E}^{\mathbb{P}}[E'] > 1$, contradicting E' being an e-variable – so exact e-variables are always admissible.

Step 3. Define $\mathbb{S}$ via $\mathrm{d}\mathbb{S}/\mathrm{d}\mathbb{P} = E$; since $E \geq 0$ is exact, $\mathbb{S}$ is a genuine probability measure absolutely continuous w.r.t. $\mathbb{P}$. ∎

2.5 Admissible E-Variables IV

2.5.2 Characterization of Admissibility (General $\mathcal{P}$)

For $A \in \mathcal{F}$ and $\delta > 0$, write $\mathcal{P}_A = \{\mathbb{P} \in \mathcal{P} : \mathbb{P}(A) > 0\}$ and $\mathcal{P}_A^\delta = \{\mathbb{P} \in \mathcal{P} : \mathbb{P}(A) > \delta\}$.

Proposition 2.13 (necessary condition). If E is admissible for $\mathcal{P}$, then for any $A \in \mathcal{F}$ with $\mathcal{P}_A \neq \varnothing$, $\sup_{\mathbb{P} \in \mathcal{P}_A} \mathbb{E}^{\mathbb{P}}[E] = 1$.

Theorem 2.14 (full characterization). $E : \Omega \to [0, \infty]$ is an admissible e-variable for $\mathcal{P}$ if and only if

$$\liminf_{\delta \downarrow 0} \inf_{\mathbb{P} \in \mathcal{P}_A^\delta} \frac{1 - \mathbb{E}^{\mathbb{P}}[E]}{\delta} = 0 \qquad \text{for all } A \in \mathcal{F}.$$

- $\mathcal{P}_A$: the candidate laws that give event A positive probability; $\mathcal{P}_A^\delta$: those giving it probability above the threshold δ.
- Theorem 2.14's condition says: for every event A, as we shrink the required probability δ of A toward 0, we can always find candidate laws under which E gets arbitrarily close to being exact *relative to how much probability mass* A carries.

2.5 Admissible E-Variables V

Corollary 2.15: Easier Sufficient Conditions

An e-variable E for $\mathcal{P}$ is admissible if any one of:

(a) for each $A \in \mathcal{F}$ with $\mathcal{P}_A \neq \varnothing$, there exists $\mathbb{P} \in \mathcal{P}_A$ with $\mathbb{E}^{\mathbb{P}}[E] = 1$ (**strong admissibility**);

(b) for each $\mathbb{P}' \in \mathcal{P}$, there exists $\mathbb{P} \in \mathcal{P}$ with $\mathbb{P}' \ll \mathbb{P}$ and $\mathbb{E}^{\mathbb{P}}[E] = 1$;

(c) E is exact.

Strong Admissibility

Condition (a) is called **strong admissibility**. It is strictly stronger than plain admissibility in general (for infinite $\mathcal{P}$, an admissible E need not attain $\mathbb{E}^{\mathbb{P}}[E] = 1$ for *any* single $\mathbb{P}$). But when $\mathcal{P}$ is **finite**, Proposition 2.16 shows strong admissibility and admissibility *coincide*.

2.5 Admissible E-Variables VI

2.5.3 Admissible E-Variables on Product Spaces

Let $\mathcal{P}'$ be probabilities on another space, and $\mathcal{P} \times \mathcal{P}' := \{\mathbb{P} \times \mathbb{P}' : \mathbb{P} \in \mathcal{P}, \mathbb{P}' \in \mathcal{P}'\}$. For E on Ω and E' on Ω', the product $E \times E'$ (defined by $(\omega, \omega') \mapsto E(\omega)E'(\omega')$) is an e-variable for $\mathcal{P} \times \mathcal{P}'$, because $\mathbb{E}^{\mathbb{P} \times \mathbb{P}'}[E \times E'] = \mathbb{E}^{\mathbb{P}}[E]\, \mathbb{E}^{\mathbb{P}'}[E'] \leq 1$.

Proposition 2.17

If E and E' are *strongly* admissible for $\mathcal{P}$ and $\mathcal{P}'$ respectively, then $E \times E'$ is strongly admissible for $\mathcal{P} \times \mathcal{P}'$. *(Proof idea: any product event B contains a rectangle $A \times A'$ of positive product measure; apply strong admissibility of E, E' on A, A' separately and combine.)*

Practical Takeaway

Multiplying independent e-values for independent sub-problems (e.g. n i.i.d. coin flips, each contributing its own e-variable) preserves admissibility, as long as $\mathcal{P}$ and $\mathcal{P}'$ are finite (so strong admissibility applies via Proposition 2.16).

Python: Checking Admissibility on the Coin (Simple Null)

```
import numpy as np
from scipy.stats import binom

n, x, theta0 = 20, 14, 0.5
# Simple null {theta0}: Prop. 2.12 says E is admissible iff EXACT,
# i.e. E_theta0[E] == 1.
def e_value_lr(k, theta1):
    loglik = lambda th: k*np.log(th) + (n-k)*np.log(1-th)
    return np.exp(loglik(theta1) - loglik(theta0))

theta1 = x / n                     # 0.7, MLE alternative used throughout
k = np.arange(n + 1)
pmf0 = binom.pmf(k, n, theta0)
E_k = e_value_lr(k, theta1)
mean_E = np.sum(pmf0 * E_k)
print(f"E_theta0[E] = {mean_E:.6f}")   # 1.000000 => exact => admissible

# Contrast: mean-based e-variable from Sec. 2.6.3, (1-lam)+lam*Z/mu,
# lam=1, Z=heads, mu=10 -- admissible via Corollary 2.15(b) though not
# exact in general (only exact at the boundary case Z=mu a.s.)
mu = n * theta0
mean_E_mean = np.sum(pmf0 * (k / mu))
print(f"E_theta0[Z/mu] = {mean_E_mean:.6f}")   # = 1.0 for this theta0
```

Outline

2.6.1 Monotone Likelihood Ratio I

Many parametric families used in practice – including the Binomial family behind our coin example – have the property that likelihood ratios are monotone in the data. This lets us build e-variables for one-sided hypotheses almost for free.

Proposition 2.18

Let $(\mathbb{P}_\theta)_{\theta\in\Theta}$ have densities $(p_\theta)_\theta$ with **monotone likelihood ratio**: for $\theta < \theta'$, $p_{\theta'}(z)/p_\theta(z)$ is increasing (or decreasing) in z. Consider testing $\mathcal{P} = \{\mathbb{P}_\theta : \theta \le \theta_0\}$ against $\mathcal{Q} = \{\mathbb{P}_\theta : \theta \ge \theta_1\}$, $\theta_1 > \theta_0$. Then for any $\theta' > \theta_0$,

$$E = \frac{p_{\theta'}(Z)}{p_{\theta_0}(Z)}$$

is an e-variable for $\mathcal{P}$.

- Monotone likelihood ratio (MLR): larger data values z favor larger θ – a very common "well-behaved" pattern (satisfied by the Binomial, Normal, Poisson, and other exponential families).
- The claim covers our whole running example: $E \approx 5.18$ from Section 2.1 is precisely an instance of Proposition 2.18 with $\theta_0 = 0.5$, $\theta' = \theta_1 = 0.7$, since the Binomial family has MLR.

Proof Sketch (via Chebyshev's Association Inequality)

Step 1. Chebyshev's association inequality: for increasing f, g and any random variable X, $\mathbb{E}[f(X)g(X)] \geq \mathbb{E}[f(X)]\mathbb{E}[g(X)]$.

Step 2. For $\theta \leq \theta_0 < \theta'$, write

$$1 = \mathbb{E}^{\theta_0}\left[\frac{p_{\theta'}(Z)}{p_{\theta_0}(Z)}\right] = \mathbb{E}^{\theta}\left[\frac{p_{\theta'}(Z)}{p_{\theta_0}(Z)} \cdot \frac{p_{\theta_0}(Z)}{p_{\theta}(Z)}\right] \geq \mathbb{E}^{\theta}\left[\frac{p_{\theta'}(Z)}{p_{\theta_0}(Z)}\right]\mathbb{E}^{\theta}\left[\frac{p_{\theta_0}(Z)}{p_{\theta}(Z)}\right] = \mathbb{E}^{\theta}\left[\frac{p_{\theta'}(Z)}{p_{\theta_0}(Z)}\right],$$

using Step 1's inequality (both ratios above are increasing in z under the MLR assumption) and $\mathbb{E}^{\theta}[p_{\theta_0}(Z)/p_{\theta}(Z)] = 1$ (a change-of-measure identity). This gives $\mathbb{E}^{\theta}[E] \leq 1$ for all $\theta \leq \theta_0$, as required.
∎

Optimality Preview

Section 6.7.1 will show this e-variable is actually *log-optimal* against $\mathbb{Q} = \mathbb{P}_{\theta_1}$ – simply because $p_{\theta_1}/p_{\theta_0}$ is itself an e-variable for $\mathcal{P}$, the best possible choice against that specific alternative.

2.6.2–2.6.7 Further Examples I

The book gives several more constructions of e-values for different testing problems, several of which will recur (and be shown optimal) in later chapters.

2.6.2 Testing Symmetry

$\mathcal{P} = \{\mathbb{P} : Z \overset{d}{=} -Z \text{ under } \mathbb{P}\}$ (symmetric around 0). For a single observation with alternative density q, using $p^*(z) = \frac{1}{2}(q(z) + q(-z))\mathbf{1}_{\{q(z)>0\}}$,

$$E = \frac{2q(Z)}{q(Z) + q(-Z)}\mathbf{1}_{\{q(Z)>0\}}$$

is an exact, admissible e-variable (Eq. 2.8). For n i.i.d. symmetric data $X_1, \ldots, X_n$, natural families of e-variables include

$$E_\lambda = \prod_{i=1}^{n} \frac{2\exp(\lambda X_i)}{\exp(\lambda X_i) + \exp(-\lambda X_i)}, \qquad E_p = \frac{p^k(1-p)^{n-k}}{2^{-n}},$$

where k is the number of positive X_i's (using that signs are i.i.d. $\mathrm{Uniform}\{-1, 1\}$ under $\mathcal{P}$), plus signed-rank e-variables E_β based on the Wilcoxon statistic.

2.6.3 Testing the Mean (and Variance)

For nonnegative Z and $\mathcal{P} = \{\mathbb{P} : \mathbb{E}^{\mathbb{P}}[Z] \leq \mu\}$, both 1 and Z/μ are e-variables, hence so is any mixture $(1-\lambda) + \lambda\, Z/\mu$, $\lambda \in [0,1]$ (Eq. 2.9) – admissible by Corollary 2.15(b) despite not being exact. Adding a variance bound $\mathrm{var}^{\mathbb{P}}(Z) \leq \sigma^2$ gives, via $\mathbb{E}^{\mathbb{P}}[Z^2] \leq \mu^2 + \sigma^2$, the further e-variable $(1-\lambda) + \lambda\, Z^2/(\mu^2+\sigma^2)$ (Eq. 2.10). For dependent data $(X_1, \ldots, X_n)$ with common marginal mean $\leq \mu$, $(X_1 + \cdots + X_n)/(n\mu)$ is an e-variable *regardless of the dependence structure* – no independence assumption needed at all.

Running Example: The Mean-Based E-Variable for the Coin

View each flip as $X_i \in \{0,1\}$ with $\mathbb{P}(X_i = 1) = \theta$; under $H_0 : \theta \leq 0.5$, the sum $Z = \sum_i X_i$ has mean at most $\mu = n \times 0.5 = 10$. With observed $Z = 14$, the e-variable $(1-\lambda) + \lambda Z/\mu$ is maximized at $\lambda = 1$:

$$E_{\text{mean}} = Z/\mu = 14/10 = 1.4.$$

Compare to the parametric likelihood-ratio e-value ≈ 5.18 from Section 2.1: the distribution-free, mean-only e-variable is *much weaker* evidence, illustrating the price paid for making fewer assumptions about the data-generating process.

2.6.2–2.6.7 Further Examples V

2.6.4 Testing Sub-Gaussian Means

$\mathcal{P} = \{\mathbb{P} : \mathbb{E}^{\mathbb{P}}[e^{\lambda Z - \lambda^2/2}] \leq 1 \ \forall \lambda \geq 0\}$ (right tail lighter than a unit Gaussian, mean ≤ 0). The family $\{e^{\lambda Z - \lambda^2/2} : \lambda \geq 0\}$, and any mixture thereof, gives e-variables for $\mathcal{P}$. A closely related null $\mathcal{P}^0$ (sub-Gaussian on *both* tails, mean exactly 0) additionally admits Z^2 as an e-variable, because every $\mathbb{P} \in \mathcal{P}^0$ has variance ≤ 1.

2.6.5 Constrained Hypotheses

Generalizing 2.6.3–2.6.4: for a constraint set $\Psi \subseteq \mathcal{L}$ (real measurable functions), define $\mathcal{P} = \{\mathbb{P} \in \mathcal{M}_1 : \int f \, \mathrm{d}\mathbb{P} \leq 0 \ \forall f \in \Psi\}$ (Eq. 2.13). For finite $\Psi = \{g_1, \ldots, g_K\}$, *every* e-variable for $\mathcal{P}$ is (quasi-surely) dominated by $1 + \sum_{k=1}^{K} \pi_k g_k$ for some $\pi \in \mathbb{R}_+^K$ – a clean, complete description of all e-variables in terms of the defining constraints.

2.6.2–2.6.7 Further Examples VI

2.6.6 Testing Likelihood Ratio Bounds

$\mathcal{P} = \{\mathbb{P} : \mathrm{d}\mathbb{P}/\mathrm{d}\mathbb{P}_0 \leq \gamma\}$ against $\mathbb{Q} \ll \mathbb{P}_0$ (closeness to a benchmark law $\mathbb{P}_0$, in likelihood-ratio terms). The naive $E = (\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}_0)/\gamma$ is an e-variable; it can be sharpened via quantile truncation to an improved e-variable $E^* \geq E$, using a classical risk-measure identity (Ch. 16).

2.6.7 Testing Exchangeability

$\mathcal{P} = \{\mathbb{P} : Z \stackrel{d}{=} Z_\sigma \ \forall \sigma \in \mathcal{S}_n\}$, where Z_σ permutes the n coordinates. With π a uniform random permutation independent of Z and alternative density q,

$$E = \frac{q(Z)}{\frac{1}{n!}\sum_{\sigma \in \mathcal{S}_n} q(Z_\sigma)}$$

is an exact, admissible e-variable – a "soft-rank" e-value (Section 1.7).

Python: More E-Value Examples on the Coin

```python
import numpy as np

n, x, theta0 = 20, 14, 0.5
mu = n * theta0                                 # mean under H0: theta <= 0.5

# Sec. 2.6.3: distribution-free mean-based e-variable, lambda=1
E_mean = x / mu
print(f"mean-based e-value (Z/mu) = {E_mean:.3f}")      # 1.400

# Sec. 2.6.1: parametric MLR-based e-value (theta1 = MLE)
theta1 = x / n
loglik = lambda th: x*np.log(th) + (n-x)*np.log(1-th)
E_lr = np.exp(loglik(theta1) - loglik(theta0))
print(f"likelihood-ratio e-value = {E_lr:.3f}")         # 5.184

# Sec. 2.6.2 flavor: sign-based e-value, k = #heads = #"+1" signs
# (signs i.i.d. Uniform{-1,1} under H0: fair coin), candidate bias p
k, p_cand = x, 0.7
E_signs = (p_cand**k * (1 - p_cand)**(n - k)) / (2**-n)
print(f"sign-based e-value at p={p_cand} = {E_signs:.3f}")
# E_lr and E_signs coincide: both reduce to the same binomial
# likelihood-ratio construction.
```

Outline

2.7 Informal Grids for Realized E-Values I

P-values are conventionally judged against thresholds like 0.01 or 0.05. What is the analogous rule of thumb for e-values? Since e-values are generalized likelihood ratios / Bayes factors, we can borrow Jeffreys's classical grading scale for likelihood ratios.

Table 2.19: Jeffreys's Rule of Thumb, Applied to E-Values

e-value	level of evidence	Shafer's p-value
$[0, 1)$	null hypothesis is supported	$(0.25, 1]$
$(1, 3.16)$	no more than a bare mention	$(0.0577, 0.25)$
$(3.16, 10)$	substantial	$(8.3\times10^{-3}, 0.0577)$
$(10, 31.6)$	strong	$(9.4\times10^{-4}, 8.3\times10^{-3})$
$(31.6, 100)$	very strong	$(9.8\times10^{-5}, 9.4\times10^{-4})$
$(100, \infty]$	decisive	$[0, 9.8\times10^{-5})$

2.7 Informal Grids for Realized E-Values II

- "Shafer's p-value" column: the range of p mapping to each e-value band under the calibrator $e = p^{-1/2} - 1$ (Eq. 2.4), included purely for comparison – it is *not* a claim that e-values and p-values correspond one-to-one in general.
- These thresholds ($e > 3.16$, $e > 10$, etc.) work best for e-values obtained "naturally" (likelihood ratios or their generalizations), **not** for all-or-nothing e-variables like $E = \mathbf{1}_{\{P \leq \alpha\}}/\alpha$, where $E = 10$ is just $P \leq 0.1$ – a very different notion of "strength."

2.7 Informal Grids for Realized E-Values III

Running Example: Grading Our Coin E-Value

Our natural likelihood-ratio e-value $E \approx 5.18$ falls in the range $(3.16, 10)$: **"substantial"** evidence against fairness by Jeffreys's scale – consistent with (though not numerically identical to) the one-sided p-value ≈ 0.058, which sits just above the conventional 0.05 cutoff.

E-Values and P-Values Can Point in Different Directions!

A cautionary example (two-sided normal test, $Z \sim N(\mu, 1)$): the classical p-value $P = 1 - \Phi(Z)$ does *not* depend on which alternative $\delta = \mu$ you had in mind (it is uniformly most powerful), but the likelihood-ratio e-value $E = \exp(\delta Z - \delta^2/2)$ depends strongly on δ. With $\delta = 5$ and $Z = 2$: $P \approx 0.023$ (looks like evidence *against* H_0), yet $E = \exp(10 - 12.5) = e^{-2.5} \approx 0.082$ (evidence *for* H_0, since $E < 1$)! **Lesson**: natural e-values and p-values need not agree in direction, because they encode different notions of evidence – this is another reason not to compare them naively.

Outline

What We Learned

- **Markov's inequality** (Prop. 2.1): $1/E$ is always a p-variable – the universal, if conservative, bridge from any e-value to a level-α test.
- Knowing the **exact null distribution** of E (Sec. 2.2) gives sharper thresholds than $1/\alpha$.
- **Calibrators** (Sec. 2.3): the e-to-p direction has a *unique* admissible calibrator $\min(1, 1/t)$; the p-to-e direction admits a rich family.
- **External randomization** (Sec. 2.4) strictly improves on Markov's inequality (Theorem 2.9).
- **Admissible e-variables** (Sec. 2.5) cannot be enlarged; for simple nulls they are exactly the likelihood ratios.
- Sec. 2.6 catalogued e-variables for MLR families, symmetry, means/variances, sub-Gaussian means, constrained hypotheses, likelihood-ratio bounds, and exchangeability.
- Sec. 2.7's Jeffreys-style grid interprets a realized e-value's strength – but e-values and p-values need not agree in *direction*.
- **Running example**: our coin ($n = 20$, 14 heads) gave a likelihood-ratio e-value ≈ 5.18 and a p-value ≈ 0.058, calibrated between both directions throughout.